

11.10 Homework: Integration Review (with calculator)

1. A botanist is conducting an experiment which studies the growth of plants. The following table shows the number of days, d , that the experiment has been running and the average height, h cm, of the plants on each of those days.

d	2	5	13	24	33	37	42
h	10	16	30	59	76	79	82

- (a) The regression line of h on d for this data can be written in the form $h = ad + b$. Find the value of a and the value of b . [2]
- (b) Write down the value of the Pearson's product-moment correlation coefficient, r . [1]
- (c) Use your regression line to estimate the average height of the plants when the experiment has been running for 20 days. [2]

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2. In this question, give all answers correct to two decimal places.

Sam invests \$1700 in a savings account that pays a nominal annual rate of interest of 2.74%, compounded half-yearly. Sam makes no further payments to, or withdrawals from, this account.

- (a) Find the amount that Sam will have in his account after 10 years. [3]

David also invests \$1700 in a savings account that pays an annual rate of interest of $r\%$, compounded yearly. David makes no further payments or withdrawals from this account.

- (b) Find the value of r required so that the amount in David's account after 10 years will be equal to the amount in Sam's account. [2]

- (c) Find the interest David will earn over the 10 years. [1]

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3. [Maximum mark: 5]

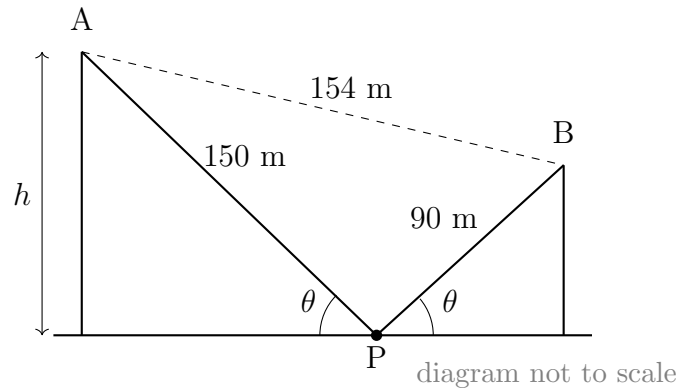
A geometric sequence has a first term of 50 and a fourth term of 86.4.

The sum of the first n terms of the sequence is S_n .

Find the smallest value of n such that $S_n > 33\,500$.

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4. The following diagram shows two buildings situated on level ground. From point P on the ground directly between the two buildings, the angle of elevation to the top of each building is θ .



The distance from point P to point A is 150 m.

The distance from point P to point B is 90 m.

The distance between A and B is 154 m.

(a) Find the measure of $\angle APB$. [3]

(b) Find the height, h , of the taller building. [3]

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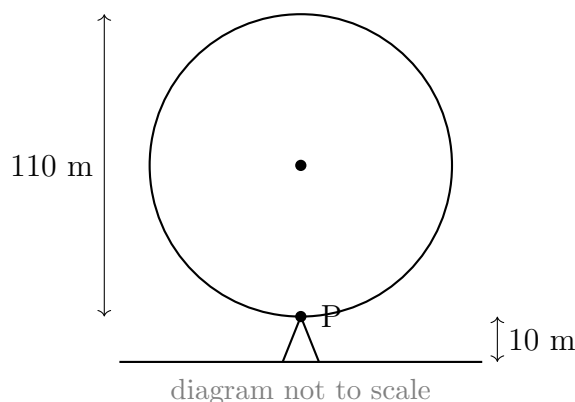
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5. [Maximum mark: 5]

A Ferris wheel with diameter 110 metres rotates at a constant speed. The lowest point on the wheel is 10 metres above the ground. P is a point on the wheel. The wheel starts moving with P at the lowest point and completes one revolution in 20 minutes.



The height, h metres, of P above the ground after t minutes is given by $h(t) = a \cos(bt) + c$, where $a, b, c \in \mathbb{R}$.

Find the values of a , b and c .

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6. [Maximum mark: 5]

The amount of a drug, in milligrams (mg), in a patient's body can be modelled by the function $A(t) = 500e^{-kt}$, where k is a positive constant and t is the time in hours after the initial dose is given.

(a) Write down the amount of the drug in the patient's body when $t = 0$. [1]

After three hours, the amount of the drug in the patient's body has decreased to 280 mg.

(b) Find the value of k . [2]

The second dose is given T hours after the initial dose, when the amount of the drug in the patient's body is 140 mg.

(c) Find the value of T . [2]

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7. [Maximum mark: 7]

All living plants contain an isotope of carbon called carbon-14. When a plant dies, the isotope decays so that the amount of carbon-14 present decreases. The amount, A , of carbon-14 present t years after death can be modelled by $A = A_0 e^{-kt}$ where $t \geq 0$ and A_0, k are positive constants.

At the time of death, a plant is defined to have 100 units of carbon-14.

(a) Show that $A_0 = 100$. [1]

The time taken for half the original amount of carbon-14 to decay is known to be 5730 years.

(b) Show that $k = \frac{\ln 2}{5730}$. [3]

(c) Find, correct to the nearest 10 years, the time taken after the plant's death for 25% of the carbon-14 to decay. [3]

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8. [Maximum mark: 7]

A particle moves in a straight line such that its velocity, $v \text{ m s}^{-1}$, at time t seconds is given by

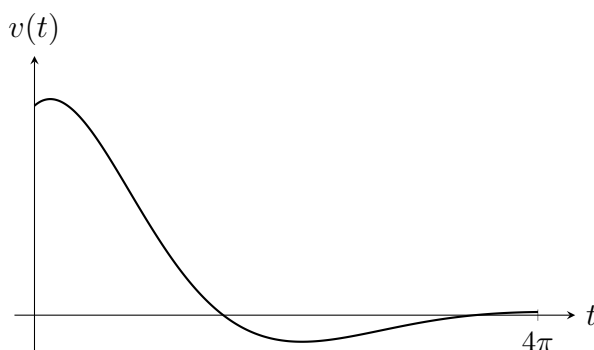
$$v = \frac{(t^2 + 1) \cos t}{4}, \quad 0 \leq t \leq 3.$$

- Determine when the particle changes its direction of motion. [2]
- Find the times when the particle's acceleration is -1.9 m s^{-2} . [3]
- Find the particle's acceleration when its speed is at its greatest. [2]

9. A particle moves in a straight line such that its velocity, $v \text{ m s}^{-1}$, at time t seconds is given by

$$v(t) = 4e^{-t/3} \cos\left(\frac{t}{2} - \frac{\pi}{4}\right), \quad \text{for } 0 \leq t \leq 4\pi.$$

The graph of v is shown in the following diagram.



Let t_1 be the first time when the particle's *acceleration* is zero.

- (a) Find the value of t_1 . [2]

Let t_2 be the *second* time when the particle is instantaneously at rest.

- (b) Find the value of t_2 . [2]

- (c) Find the distance travelled by the particle between $t = t_1$ and $t = t_2$. [2]

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